

1.Find the Area of a Rectangle Using Classes in Python

a)class rectangle():

```
def __init__(self, rectbreadth, rectlength):  
    self.rectbreadth = rectbreadth  
    self.rectlength = rectlength
```

```
def areaofRect(self):  
    return self.rectbreadth*self.rectlength
```

```
rectlength = 31  
rectbreadth = 19
```

```
rectObj = rectangle(rectlength, rectbreadth)  
print("The area of the rectangle with the given sides", rectlength, ',', rectbreadth, '=', rectObj.areaofRect())
```

Output: The area of the rectangle with the given sides 31 , 19 = 589

b)Using Classes(User Input separated by spaces)

class rectangle():

```
def __init__(self, rectbreadth, rectlength):  
    self.rectbreadth = rectbreadth  
    self.rectlength = rectlength
```

```
def areaofRect(self):  
    return self.rectbreadth*self.rectlength
```

```
rectlength, rectbreadth = map(int, input('Enter the length and breadth of the rectangle separated by spaces = ').split())  
rectObj=rectangle(rectlength, rectbreadth)  
print("The area of the rectangle with the given sides", rectlength, ',', rectbreadth, '=', rectObj.areaofRect())
```

Output:

Enter the length and breadth of the rectangle separated by spaces = 7 9
The area of the rectangle with the given sides 7 , 9 = 63

c)Using Classes(User Input separated by newline)

```
class rectangle():

    def __init__(self, rectbreadth, rectlength):
        self.rectbreadth = rectbreadth
        self.rectlength = rectlength

    def areaofRect(self):
        return self.rectbreadth*self.rectlength

rectlength = int(input('Enter some random length of the rectangle = '))
rectbreadth = int(input('Enter some random breadth of the rectangle = '))

rectObj = rectangle(rectlength, rectbreadth)
print("The area of the rectangle with the given sides", rectlength, ',', rectbreadth, '=', rectObj.areaofRect())
```

Output:

```
Enter some random length of the rectangle = 15
Enter some random breadth of the rectangle = 11
The area of the rectangle with the given sides 15 , 11 = 165
```

2. Create a Class and Compute the Area and the Perimeter of the Circle

```
import math
class circleClass():

    def __init__(self, circleradius):
        self.circleradius = circleradius

    def printArea(self):
        return math.pi*(self.circleradius**2)

    def printPerimeter(self):
        return 2*math.pi*self.circleradius

circleradius = float(input('Enter some random radius of the circle = '))
circleobj = circleClass(circleradius)

print("The Perimeter of circle with given radius", circleradius, '=', round(circleobj.printPerimeter(), 4))
```

```
print("The Area of circle with given radius", circleradius, '=', round(circleobj.printArea(), 4))
```

Output:

```
Enter some random radius of the circle = 20.98
The Perimeter of circle with given radius 20.98 = 131.8212
The Area of circle with given radius 20.98 = 1382.8047
```

3. Create a Class which Performs Basic Calculator Operations in Python

```
class calculator():

    def __init__(self, number1, number2):
        self.number1 = number1
        self.number2 = number2

    def addNumbers(self):
        return self.number1+self.number2

    def mulNumbers(self):
        return self.number1*self.number2

    def divNumbers(self):
        return self.number1/self.number2

    def subNumbers(self):
        return self.number1-self.number2

number1 = 28
number2 = 11

calobj = calculator(number1, number2)
option = 1

while option != 0:
    print("0. For Exit")
    print("1. For Addition of given two numbers", number1, ',', number2)
```

```
print("2. For Subtraction of given two numbers", number1, ',', number2)
print("3. For Multiplication of given two numbers", number1, ',', number2)
print("4. For Division of given two numbers", number1, ',', number2)
option = int(input("Enter the option/choice you want = "))
if option == 1:
    print("The additon of given numbers", number1,
          ',', number2, '=', calobj.addNumbers())
elif option == 2:
    print("The difference of given numbers", number1,
          ',', number2, '=', calobj.subNumbers())
elif option == 3:
    print("The product of given numbers", number1,
          ',', number2, '=', calobj.mulNumbers())
elif option == 4:
    print("The division of two numbers", number1,
          ',', number2, '=', calobj.divNumbers())
elif option == 0:
    print("Exit of program")
else:
    print("Enter the choice again")
print()
```

Output:

```
0. For Exit
1. For Addition of given two numbers 28 , 11
2. For Subtraction of given two numbers 28 , 11
3. For Multiplication of given two numbers 28 , 11
4. For Division of given two numbers 28 , 11
Enter the option/choice you want = 3
The product of given numbers 28 , 11 = 308
```

```
0. For Exit
1. For Addition of given two numbers 28 , 11
2. For Subtraction of given two numbers 28 , 11
3. For Multiplication of given two numbers 28 , 11
4. For Division of given two numbers 28 , 11
Enter the option/choice you want = 2
```

The difference of given numbers 28 , 11 = 17

4.Create a Class in which One Method Accepts a String from the User and Another Prints it

```
class stringClass():  
    def __init__(self):  
        self.string = ""  
  
    def getString(self):  
        self.string = input("Enter some random string = ")  
  
    def putString(self):  
        print("The string =", self.string)  
  
stringobj = stringClass()  
stringobj.getString()  
  
stringobj.putString()
```

Output:

Enter some random string = btechgeeks

The string = btechgeeks